

Studox.ai Roadmap Feature Report

Assessment, AI roadmap generation, selection, dashboard, roadmap display, courses progress, and remaining implementation work.

Area	Status	Notes
Architecture	Done	Roadmap architecture and contracts reviewed.
Frontend Assessment	Done	Questions, generate button, 3 option result cards, signup redirect flow present.
AI Generate API	Done / Risk	Backend route exists; depends on AI provider and strict JSON response.
Roadmap Selection	Partial	Backend select route exists; signup handoff from pending roadmap still needs completion.
Progress Tracking	Partial	Course continue updates progress, but task-level roadmap tracking is not complete.
Courses Sync	Pending	Courses are not fully dynamic based on selected roadmap goal yet.

Completed

- Architecture confirms a two-stage roadmap system: generate lightweight options first, then select and expand one full roadmap.
- Frontend assessment UI exists in public/app.js with questions, payload creation, generate action, fallback roadmaps, and 3 roadmap result cards.
- POST /api/roadmaps/generate exists in src/server.js and validates input before calling the configured AI provider.
- POST /api/roadmaps/select exists and can expand a selected lightweight roadmap into a full roadmap for logged-in MongoDB users.
- Roadmap model supports userId, careerGoal, summary, estimatedDurationWeeks, status, weeks, tasks, and resources.
- Dashboard and roadmap pages load active roadmap data and map roadmap weeks into the timeline UI.
- Course continue action exists and can update course progress plus advance roadmap progress in a generic way.

Partially Implemented

- Generate flow has frontend fallback, but backend still returns errors if the AI provider fails, returns invalid JSON, or does not return exactly 3 options.
- Roadmap card selection currently stores selected roadmap in localStorage and redirects to signup, but the signup-to-backend roadmap save handoff is not fully complete.
- Roadmap progress endpoint exists, but it mostly returns overall progress/modules/next milestone rather than week-task completion detail.
- Course progress updates roadmap generally, but there is no strong mapping between selected roadmap weeks and matching course modules yet.

Remaining Work

- After signup/login, read studox-pending-roadmap and call POST /api/roadmaps/select to save the selected roadmap automatically.

- Make courses dynamic from selected roadmap careerGoal and roadmap weeks/skills, for example Full Stack, AI/ML, Data Analyst, Cybersecurity, UI/UX.
- Create a real progress model for userId, roadmapId, courseId, completed lessons, completed tasks, completed projects, and updatedAt.
- Add task-level completion controls on the roadmap page, not only generic module progress.
- Calculate dashboard, roadmap, and courses progress from actual completed items instead of only generic increments.
- Add stronger backend fallback so /api/roadmaps/generate always returns 3 valid options even when AI quota or JSON format fails.

Risks

- Gemini/OpenAI quota or invalid AI JSON can stop real AI generation.
- ROADMAP_SELECTION_CONTRACT.md says /api/roadmaps/select is not present, but the code now has it. Documentation should be updated.
- ROADMAP_CONTRACT.md describes full roadmap shape, while /api/roadmaps/generate currently returns lightweight options. The docs need clearer separation.
- Some older/demo code still exists in public/app.js, so changes should stay targeted and avoid broad routing rewrites.

Recommended Next Priority

- First fix pending roadmap save after signup/login.
- Second make courses automatically adapt to selected careerGoal.
- Third implement real task/lesson progress tracking.
- Fourth update contracts and architecture docs to match the current two-stage flow.

Code Health Checked

- node --check public/app.js passed.
- node --check src/server.js passed.
- Git status was clean on main...origin/main during inspection.